COURSE OUTLINE

Gennoor Tech
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Agentic Architecture Reviewer

Turn system designs into traceable threat models and security recommendations

3 days (8 hours per day) | Advanced | Instructor-led, hands-on labs | Capstone project

About this course

Who should attend

Security architects, solution architects, application security, platform architects, engineering leads.

Prerequisites

- System design fundamentals
- Basic networking and identity concepts
- Threat modeling (STRIDE or similar)
- Python and API familiarity

What you will be able to do

- Extract components, data flows, identities, external dependencies and trust boundaries from structured architecture inputs
- Represent the system as a graph with provenance and uncertainty
- Apply a repeatable threat-modeling method such as STRIDE to modeled components and flows
- Generate architecture findings with evidence, assumptions, impact and recommendations
- Compare intended design with selected implementation evidence without over-claiming verification

Tools and technologies

Claude | Mermaid | JSON schema | Python | FastAPI | Terraform (evidence) | Kubernetes (evidence) | GitHub | pytest | OWASP

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DAY 1 3 modules

MODULE 1

Architecture inputs and a constrained format

- Mermaid flowcharts or a simple JSON architecture schema plus Markdown requirements
- Image-diagram extraction deferred; parsing is never treated as perfectly reliable
- Shared engineering principles and the AI-assisted workflow

HANDS-ON LAB
Parse the reference architecture inputs and report what could not be parsed.

DELIVERABLE
Ingestion layer.

MODULE 2

The system graph with provenance

- Components, data stores, identities, external systems, protocols, data classifications, trust boundaries
- Which source supports each edge; unknown and ambiguous relationships are first-class

HANDS-ON LAB
Define the typed system graph schema and build the graph for the reference design.

DELIVERABLE
Parsed architecture model.

MODULE 3

Requirements extraction

- Security requirements and architecture decisions as explicit statements with IDs, source references and verification status

HANDS-ON LAB
Extract requirements from the ADRs and link them to graph nodes.

DELIVERABLE
Requirement register.

DAY 2 3 modules

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MODULE 4

Method-driven threat generation

- STRIDE or a defined threat library applied to components and data flows
- The LLM suggests candidates; every candidate must link to a real modeled component or flow
- Assumptions, evidence references and a validation question per threat

HANDS-ON LAB
Generate STRIDE candidates with the agent and reject every one that does not map to the graph.

DELIVERABLE
Threat register with linked components.

MODULE 5

Architecture review rules

- Missing or ambiguous authentication, authorization boundaries, secrets handling, logging, encryption, data minimization, failure handling
- Security, privacy and resilience review

HANDS-ON LAB
Implement the review rules and run them on secure and flawed reference designs.

DELIVERABLE
Review rule engine.

MODULE 6

Evidence-graded findings

- Observed design facts, assumptions, missing information and recommended questions kept separate
- An unverified concern is never labelled a confirmed vulnerability
- Report schema

HANDS-ON LAB
Produce the review report and audit it for unsupported claims.

DELIVERABLE
Security architecture review report.

DAY 3 3 modules

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MODULE 7

Architect review and decision records

- Accept, reject, defer or request more evidence; rationale and owner captured
- Change tracking across design revisions

HANDS-ON LAB
Run a review session and record decisions against the findings.

DELIVERABLE
Decision workflow.

MODULE 8

Design-to-implementation comparison

- Comparing modeled requirements with Terraform, Kubernetes or repository evidence
- Reporting coverage and gaps, never claiming full proof of conformance

HANDS-ON LAB
Cross-check three requirements against infrastructure code and report the evidence gaps.

DELIVERABLE
Conformance coverage report.

MODULE 9

Evaluation and capstone

- Secure and intentionally flawed reference architectures; threat coverage, unsupported claims, evidence linkage, usefulness
- Audit trail, limitations and capstone demo

HANDS-ON LAB
Capstone build and demo.

DELIVERABLE
End-to-end architecture reviewer.

Capstone

Capstone project

A tool that ingests structured architecture inputs, builds a provenance-aware system graph, generates a linked threat model, produces evidence-graded findings and records architect decisions, with optional implementation cross-checks that report gaps rather than proof.

Assessment

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Assessment area	Weight	Evidence
Working functionality	25%	Demonstrable end-to-end workflow in a repeatable lab
Security and safety controls	20%	Threat model, authorization boundaries, approval gates, safe failure
Testing and evaluation	20%	Automated tests, golden dataset, measured quality and known limitations
Engineering quality	15%	Clear interfaces, error handling, reproducible setup, maintainable code
Evidence and explainability	10%	Traceable findings, sources, assumptions, logs
Documentation and presentation	10%	Architecture diagram, README, runbook, demo, lessons learned

Engineering principles used throughout

- Model output is a proposal, not a fact: every finding and generated change is validated by deterministic checks and tests.
- Authorization lives outside the model: tool access is enforced by application code, scoped credentials and policy.
- Human approval for consequential actions: deploying a mitigation, changing production configuration, rotating credentials.
- Decisions, evidence, model and tool versions, approvals and outcomes are logged without recording secrets.
- A model adapter and a tool adapter layer keep every build provider-agnostic and enterprise-portable.
- Quality is measured: false positives, false negatives, reliability, latency, cost and safe failure, not just a demo.

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